

Rayen Ben Cheikh

Full-Stack Developer | AI & Data Science Engineer

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PROFESSIONAL SUMMARY

I'm a developer with a master's degree in data science. I like understanding how things work and building solutions that solve a real problem. Through my different experiences, I've touched on several areas, and I enjoy learning quickly when a new project asks for something I don't know yet.

EDUCATION

Master's in data science (Professional) | Higher Institute of Informatics and Mathematics, Monastir

Aug 2023 – Nov 2025

Relevant coursework: Machine Learning, Big Data Technologies, Predictive Analytics, Data Preprocessing & Statistical Analysis

Bachelor's in computer science — Software Engineering & Information Systems | Higher Institute of Informatics and Mathematics, Monastir

Aug 2020 – Aug 2023

PROFESSIONAL EXPERIENCE

AI & Full-Stack Developer (Final Year master Internship) | Mobelite — MindCare AI Project

Feb 2025 – Aug 2025

Led the design and development of MindCare-AI, a mobile mental health companion application that integrates real-time physiological monitoring, emotion recognition, and an AI-driven conversational assistant.

- Built a Python-based REST API using computer vision and signal processing techniques (rPPG, FFT, filtering) to estimate heart rate from live facial video using only a smartphone camera.
- Trained emotion classification models on 16,000+ samples, achieving near-real-time inference for continuous emotional state detection.
- Developed an adaptive NLP chatbot that adjusts responses based on detected physiological and emotional context, providing personalized mental health support.
- Built a Flask-based backend to serve all AI models (rPPG, emotion recognition, chatbot) as REST API endpoints, enabling seamless communication with the mobile app.
- Integrated all AI services with a React Native (Expo) mobile frontend, MongoDB for data persistence, and a secure Node.js/Express/JWT backend.
- Followed Agile (Scrum) and CRISP-DM methodologies throughout the project lifecycle.

DevOps Engineer (Bachelor's Internship) | Mobelite

Feb 2023 – Jun 2023

Designed and implemented a hybrid cloud infrastructure solution with automated CI/CD pipelines, real-time monitoring, and containerized deployments.

- Architected a hybrid cloud solution integrating AWS (EC2, S3) with on-premises infrastructure for seamless scalability and cost optimization.
- Developed Ansible playbooks to automate system administration and deployment workflows.
- Built and maintained CI/CD pipelines using GitHub Actions; integrated Nexus for artifact management and SonarQube for automated code quality enforcement.
- Configured Grafana & Prometheus monitoring stack with proactive alerting for real-time system health tracking.
- Created local Kubernetes clusters and orchestrated containerized application deployments.

Mobile Developer (Internship) | Dot It

May 2022 – Jul 2022

- Built a React Native mobile application integrated with Firebase that allows users to browse trending films and manage a personal favorites list.

PERSONAL PROJECTS

Heart Disease Prediction Model

Machine learning classification system trained on 319,795 patient records (18 features) to predict heart disease risk.

- Implemented and benchmarked multiple algorithms (Random Forest, Logistic Regression, SVM, Gradient Boosting); achieved 90% precision on the test set.
- Applied feature scaling, missing-value handling, feature selection, and hyperparameter tuning.
- Stack: Python, Scikit-learn, Pandas, NumPy, Matplotlib/Seaborn, Jupyter Notebook

Academic Club Resource Management Platform

Full-stack web application for resource management, recruitment, and event coordination for academic clubs.

- Built recruitment portal, event scheduling system, and centralized resource management module.
- Stack: Django (Python), HTML/CSS/JavaScript, MySQL

Twitter Sentiment Analysis | US Accident EDA

- Analyzed and visualized sentiment patterns in social media data to understand public opinion trends.
- Performed exploratory data analysis on US traffic accident data to identify patterns related to road conditions, weather, and time of day.

TECHNICAL SKILLS

Languages	Python, JavaScript, Java, HTML/CSS, SQL, YAML
AI / ML	Scikit-learn, NLP, Computer Vision (rPPG), Pandas, NumPy, Matplotlib
Web / Mobile	React Native, Django, Node.js, Express, REST APIs, JWT
DevOps / Cloud	AWS (EC2, S3), Docker, Kubernetes, Ansible, GitHub Actions, Nexus, SonarQube
Monitoring	Grafana, Prometheus
Databases	MongoDB, MySQL
Tools	Git, Jupyter Notebook, Kaggle

CERTIFICATIONS

[AWS Academy Graduate — AWS Academy Cloud Foundations](#)

LANGUAGES

Arabic — Native

French — Bilingual

English — Professional